# Bistatic Radar for Target Detection

Guide :

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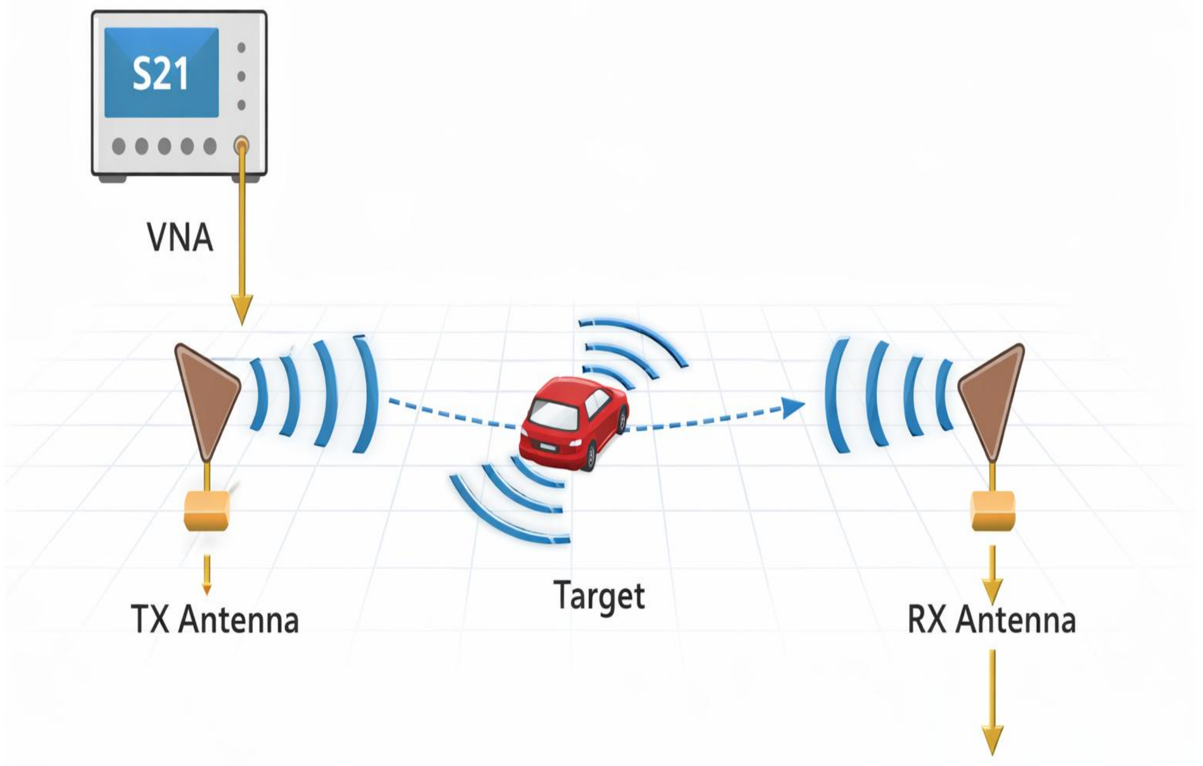
# Problem Statement

- Conventional radar systems are **expensive and complex** to implement.
- **Weak reflections from small targets** make detection difficult.
- **Strong antenna coupling and system reflections** often mask the actual target signal.
- The **target echo gets buried within the noise floor**, making reliable detection difficult.




Objective: Design a low-cost bistatic radar system for reliable target detection

# Social Impact and Applications of Low-Cost Radar Detection

## Security and Intrusion Detection

- Radar sensing systems can detect **movement or objects in restricted areas** for surveillance and monitoring.

## Industrial Object Detection

- Radar-based sensing can be used in **industrial automation to detect objects and obstacles**.

## Smart Sensing Systems

- Low-power radar can be integrated into **smart devices and robotics for environmental sensing.**

## Research and Educational Platforms

- Low-cost radar setups like this allow **students and researchers to study RF sensing and radar signal processing**.

**This project demonstrates a low-cost experimental radar platform for studying RF target detection.**

# Basic Radar Working Principle

- THE TRANSMITTING ANTENNA RADIATES AN RF SIGNAL TOWARD THE TARGET.
- THE SIGNAL REFLECTS FROM THE OBJECT SURFACE.
- THE RECEIVING ANTENNA CAPTURES THE REFLECTED SIGNAL.
- THE TIME DELAY OF THE REFLECTION IS USED TO DETERMINE THE TARGET DISTANCE.

**Distance (d)**
**d= (c×t)/2**

Where
d = distance to target
c = speed of light
t = round trip time delay

# Initial Antenna Design: CPW-Fed Ultra-Wideband Monopole

Fabricated CPW-Fed UWB Monopole Antennas

**CPW-Fed Ultra-Wideband Monopole Antenna**

• **Purpose of This Design**:

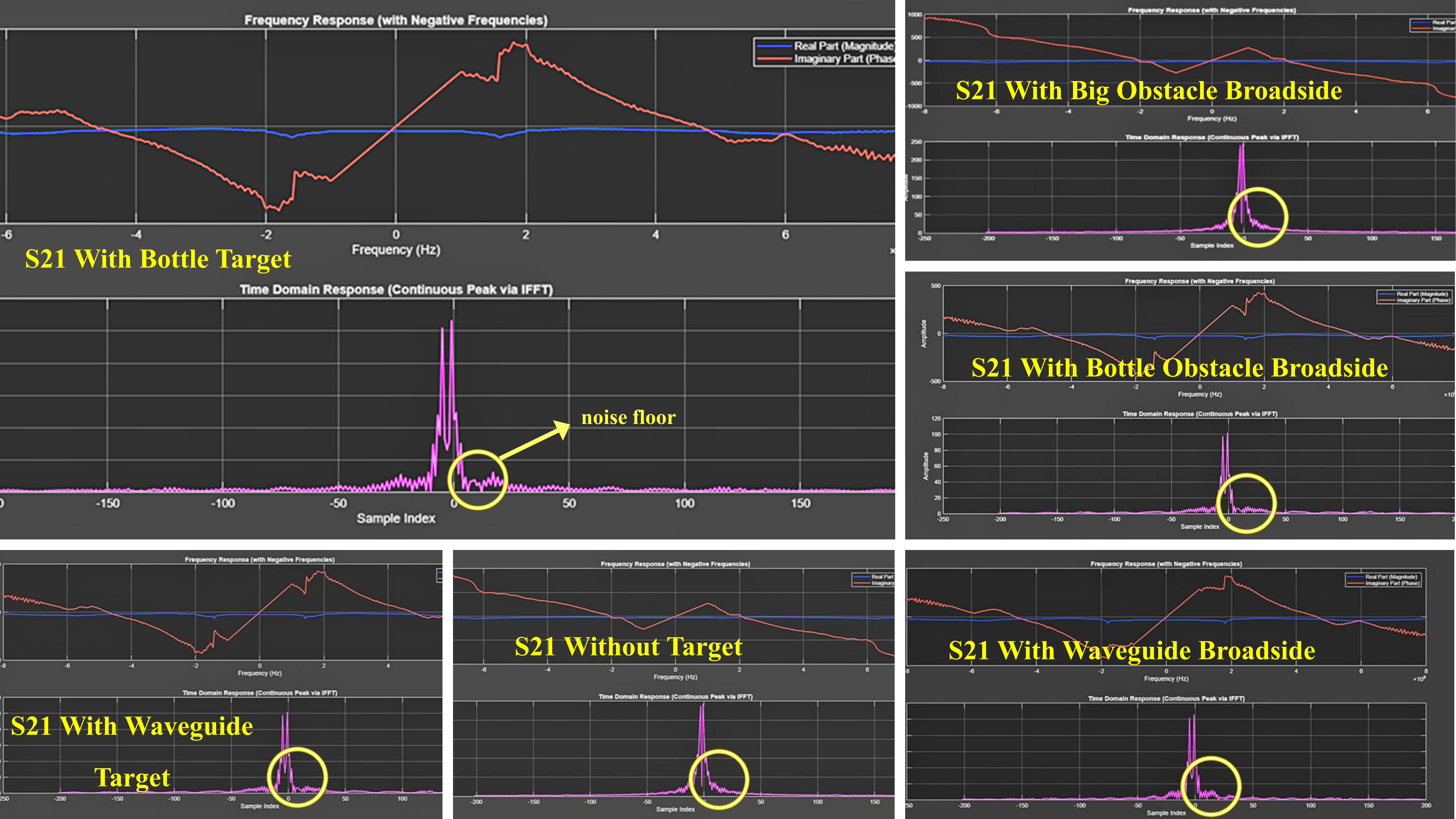
• Designed as a **CPW-fed ultra-wideband monopole antenna**.

• Intended to provide **wide bandwidth for radar sensing experiments**.

• Wide bandwidth allows observation of **multiple frequency reflections from targets**.

• **Limitations Identified :**

• **Bi-directional radiation pattern** illuminated objects behind the antenna as well.
• **Ultra-wide bandwidth introduced additional spectral noise** across the frequency range.
• This resulted in **poor target distinguishability in radar measurements**.



Frequency Response (with Negative Frequencies)
Real Part (Magnitude)
Imaginary Part (Phase)
Frequency (Hz)
S21 With Bottle Target
Time Domain Response (Continuous Peak via IFFT)
noise floor
Sample Index
S21 With Big Obstacle Broadside
S21 With Bottle Obstacle Broadside
S21 With Waveguide
Target
S21 Without Target
S21 With Waveguide Broadside

# Issues Observed with Initial Antenna

**Bi-directional radiation pattern illuminating unwanted objects**

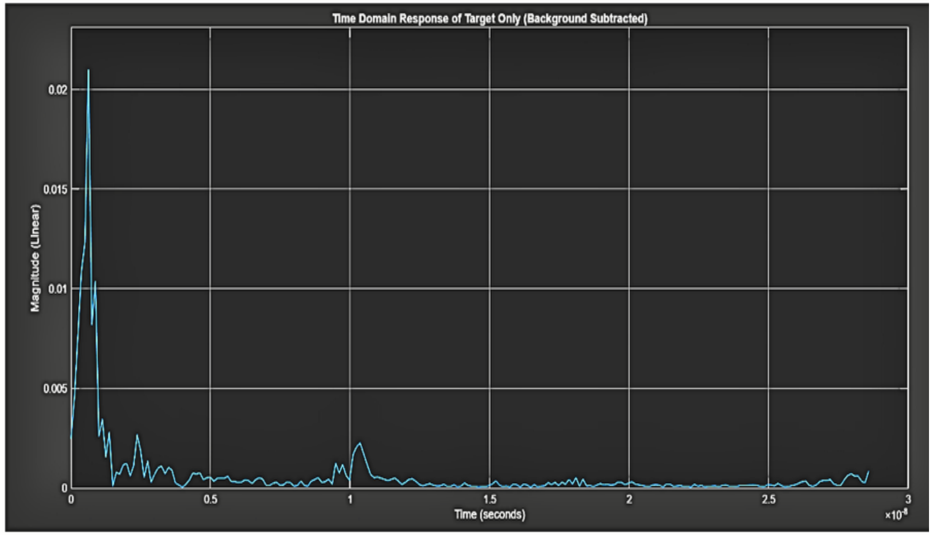
**Ultra-wide bandwidth causing additional spectral noise**

**Weak target reflections relative to system noise**

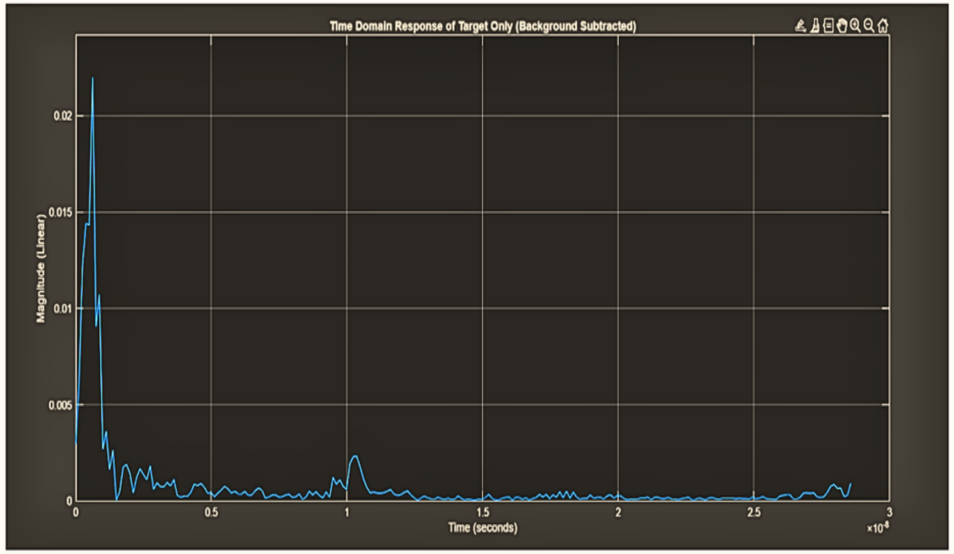
**Poor impedance matching affecting signal quality**

**Result: Antenna design unsuitable for radar imaging experiments**

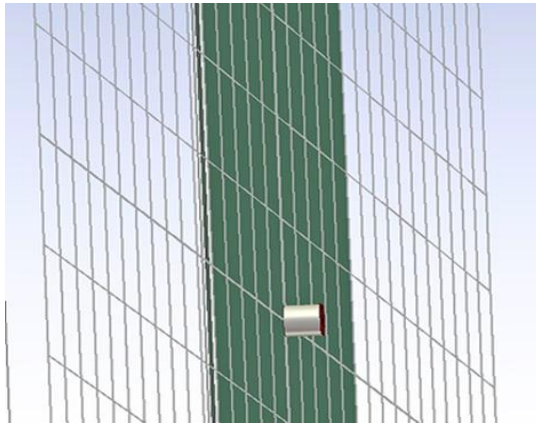
## Subtracted S21 Analysis

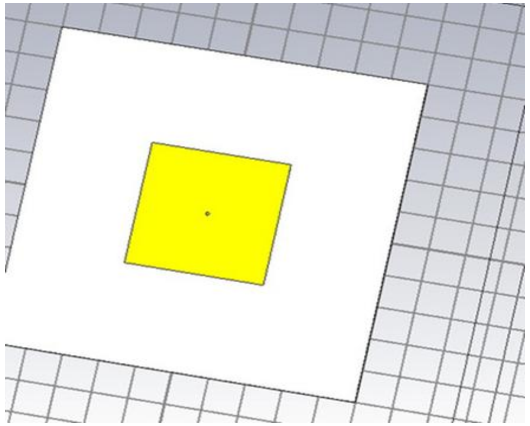



Background-subtracted S21 response shows no clear target peaks

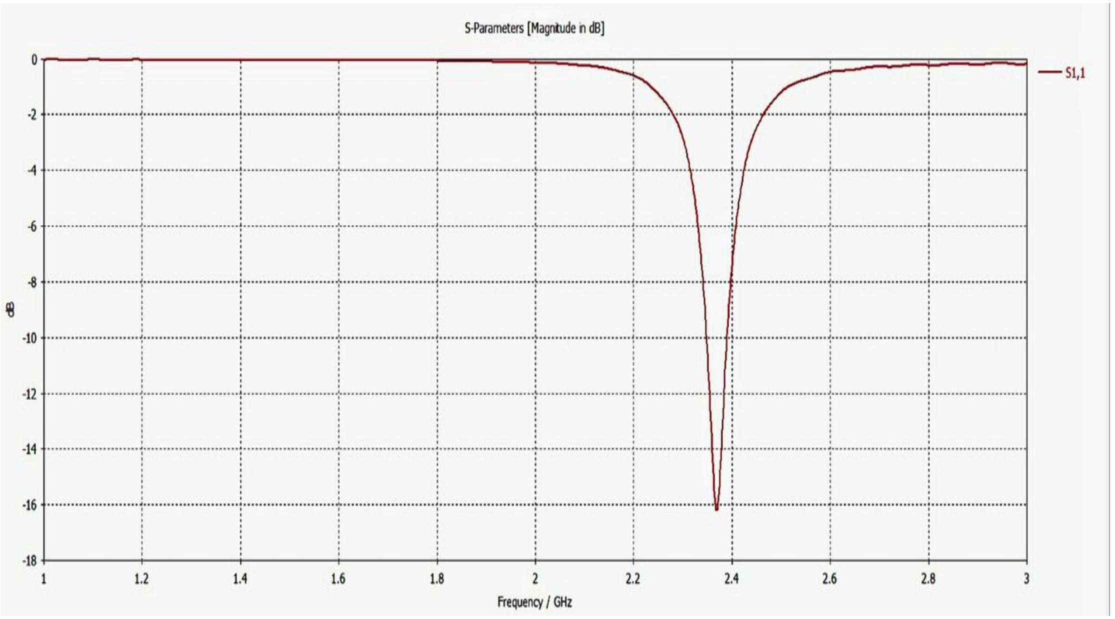


# New Design: 3 Layered Substrate Patch Antenna

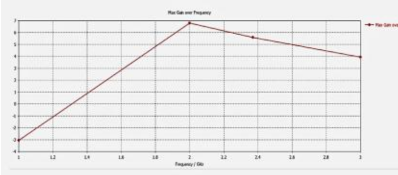




- **Design Significance :**
- We replaced the old design with an optimized 3-layered substrate patch antenna.
- **Waveform Results :**
- The first simulation achieved a good match at approx 2.45 GHz.




**Design was optimized for better impedance match & higher gain.**

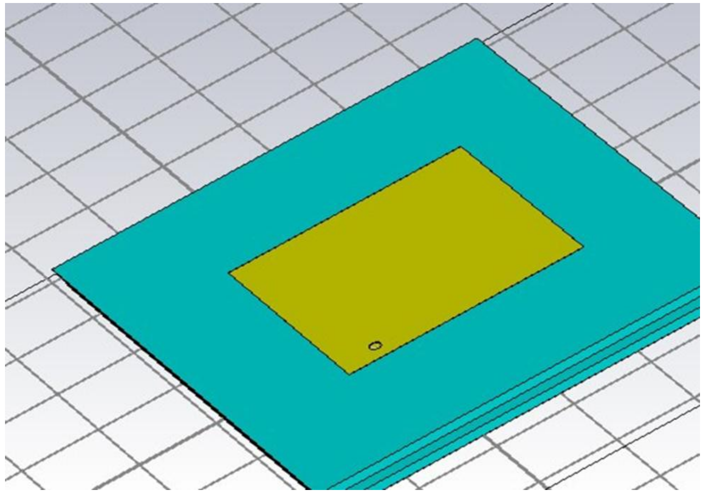


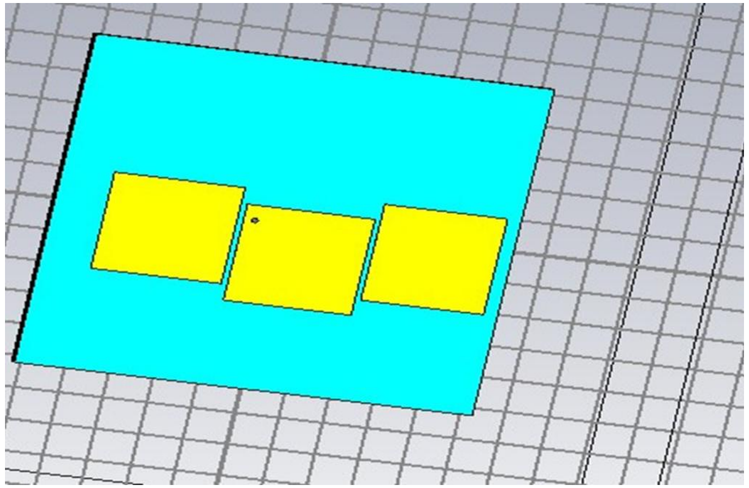
# Final Optimized Antenna Design

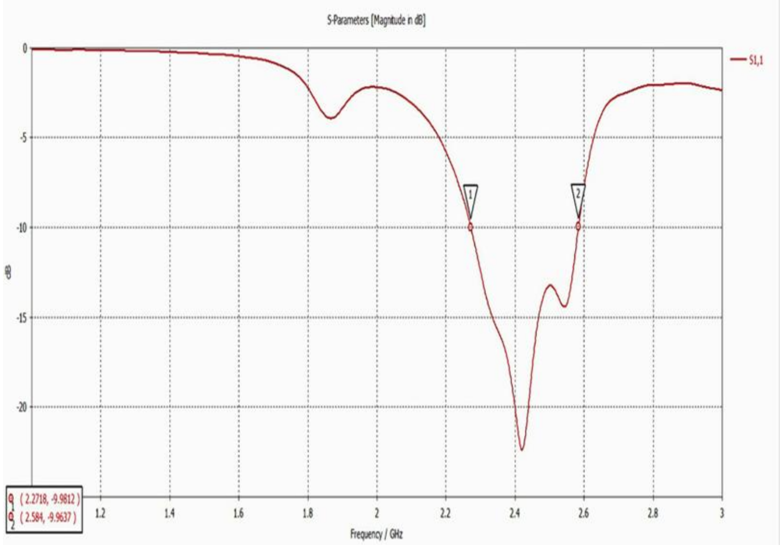
**Final Match : achieved an excellent impedance match of upto approx -27dB at resonance.**

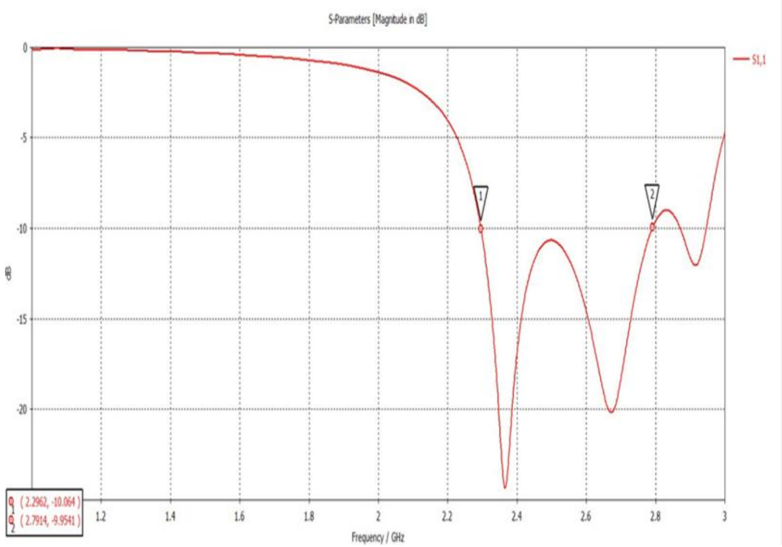
**High Gain : The design reached a high gain of approx 7 dBi.**

**Usable Bandwidth : The antenna opertaes efficiently across bandwidth greater than 500 MHz.**





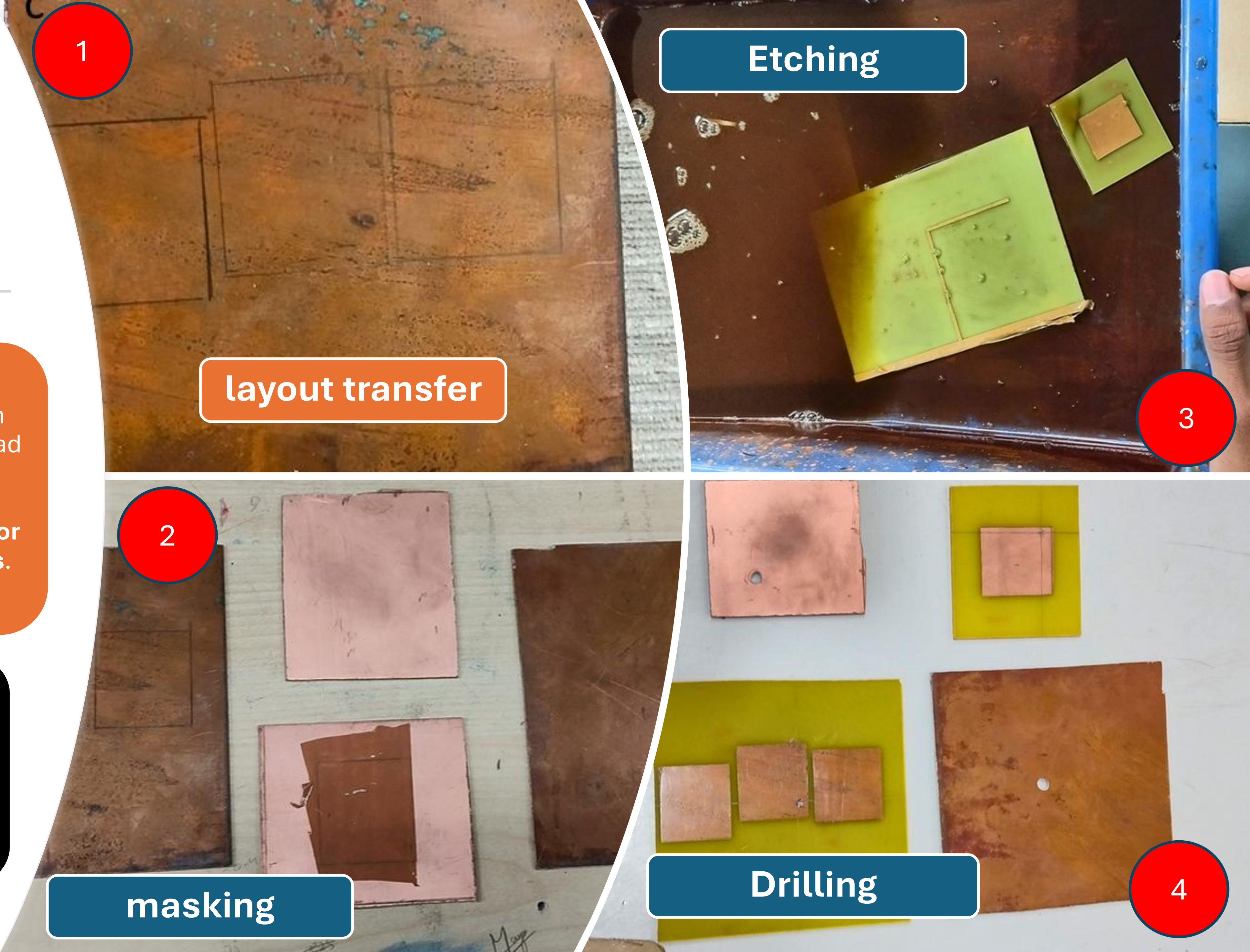


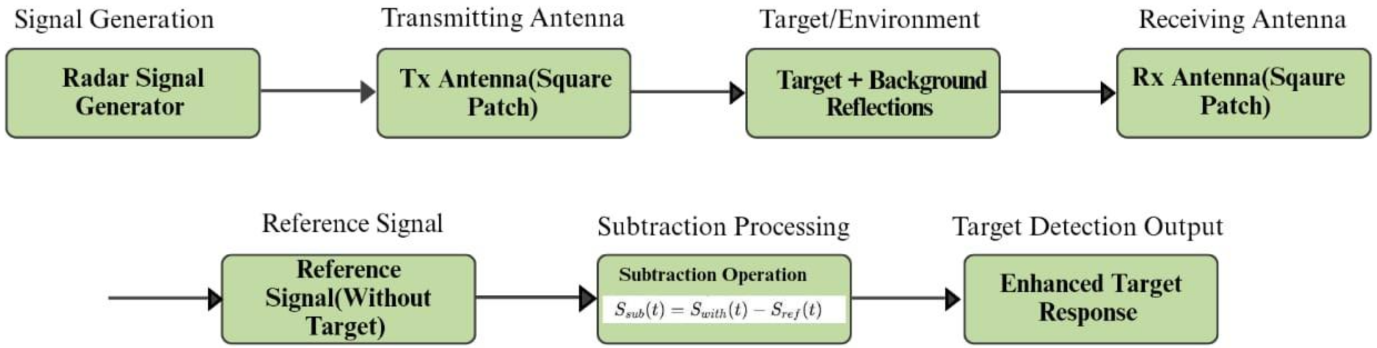
# Fabrication of Final Antenna Design

The optimized antenna design from CST was fabricated on a copper-clad substrate using **layout transfer, masking, chemical etching, and drilling**, followed by **SMA connector mounting for VNA measurements.**

**The fabricated antenna PCB matches the optimized design and is ready for VNA testing**

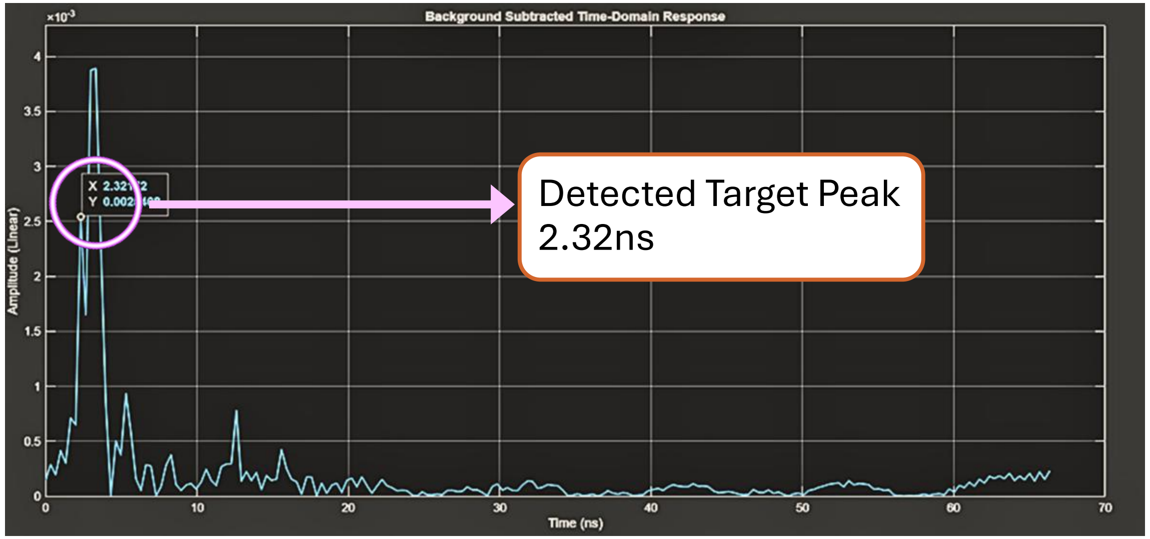


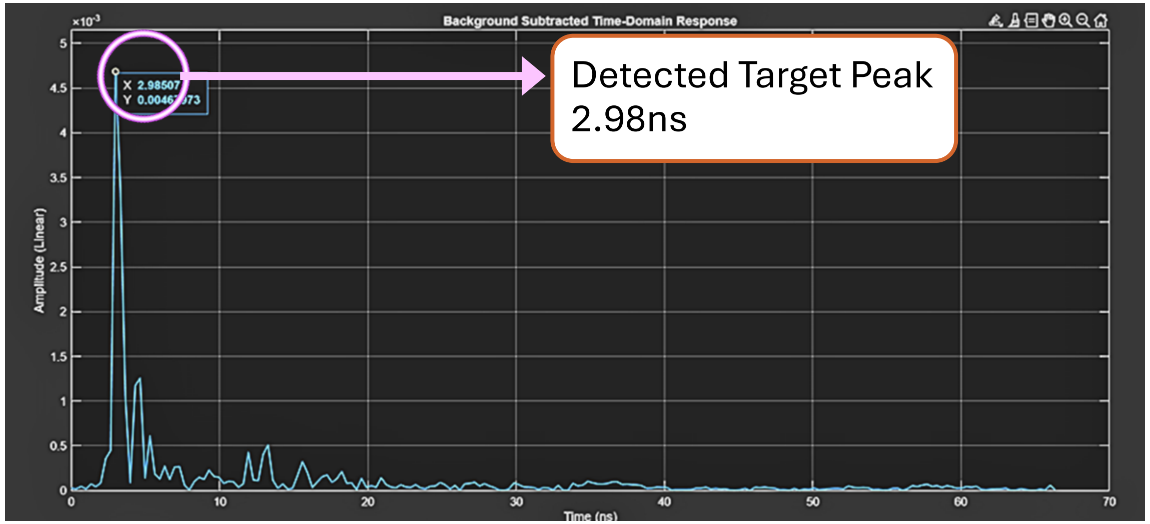
# Block Diagram



# Experimental Setup

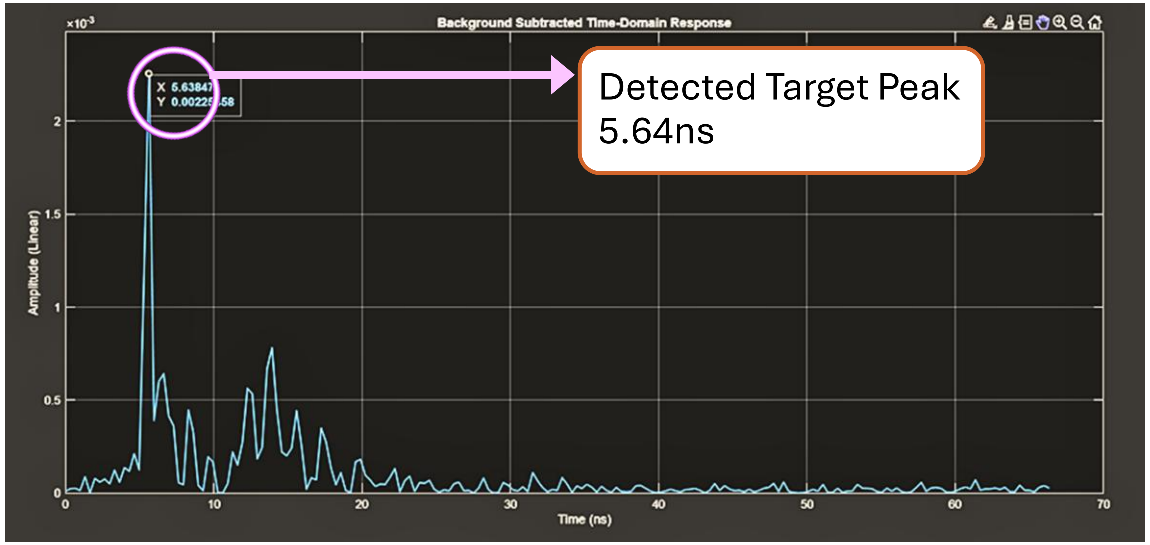
## Case 2 : S21 Response Analysis with Target at 14 inches




## Case 3: Waveguide Target Detection using Differential S21 at 16 inch

## Case 1: S21 Response Analysis with Target at 33 inches



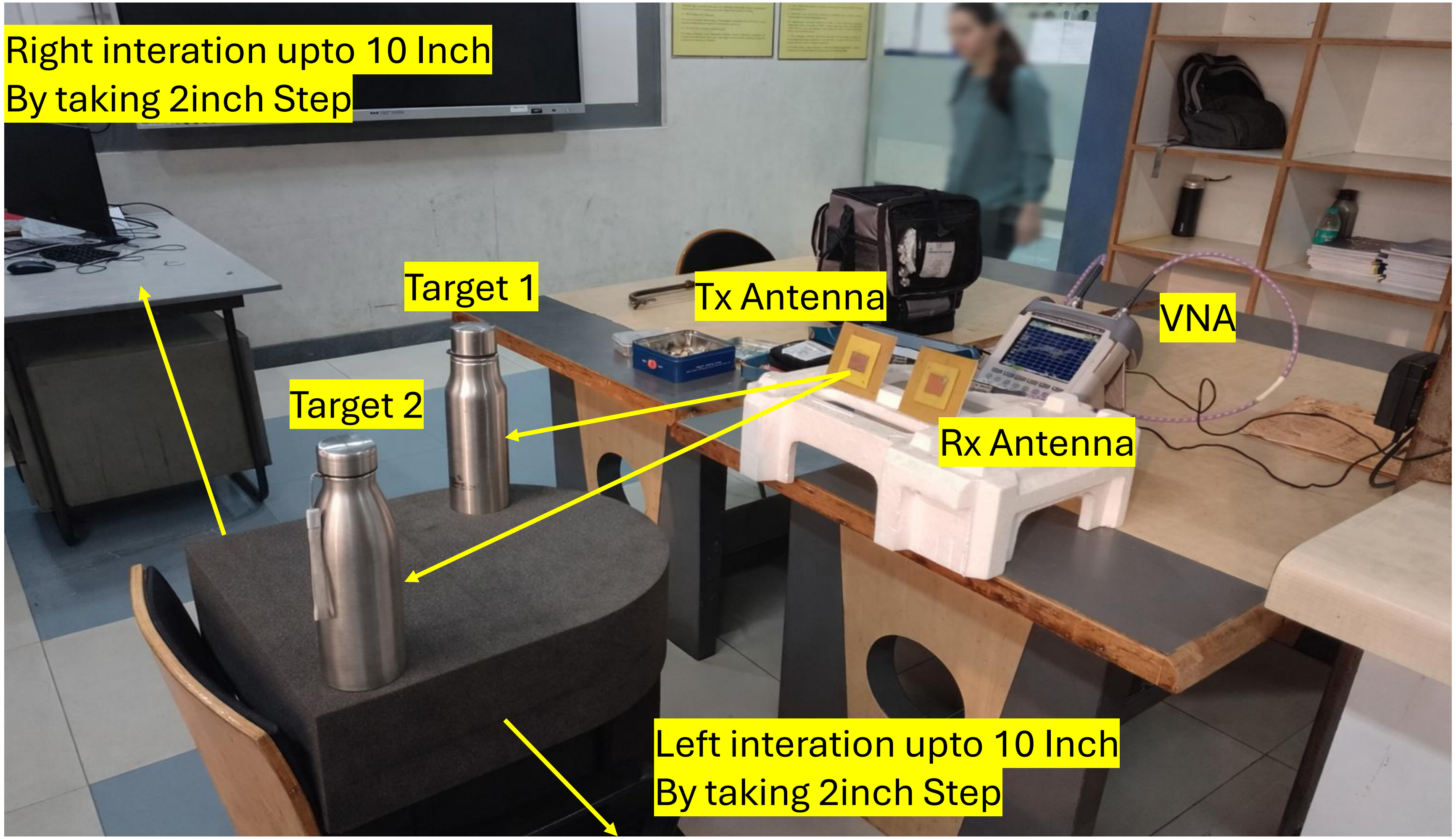

Case 1 : $$\Delta t = \frac{2 \times 0.8382}{3 \times 10^8} \approx 5.6 \text{ ns}$$

Case 2: $$\Delta t = \frac{2 \times 0.3556}{3 \times 10^8} \approx 2.37 \text{ ns}$$

Case 3: $$\Delta t = \frac{2 \times 0.4064}{3 \times 10^8} \approx 2.71 \text{ ns}$$

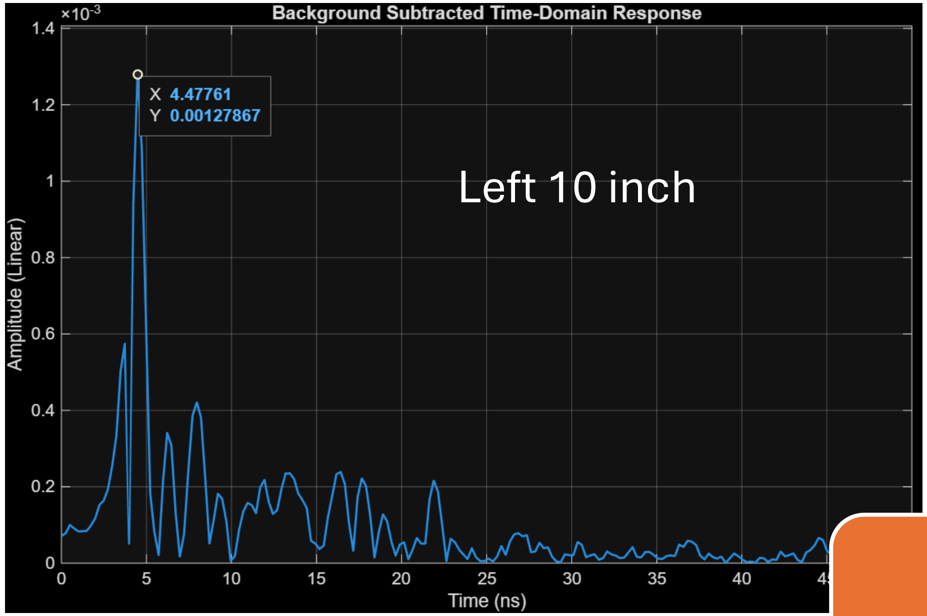
S_diff(t) = S_target(t) − S_ref(t)

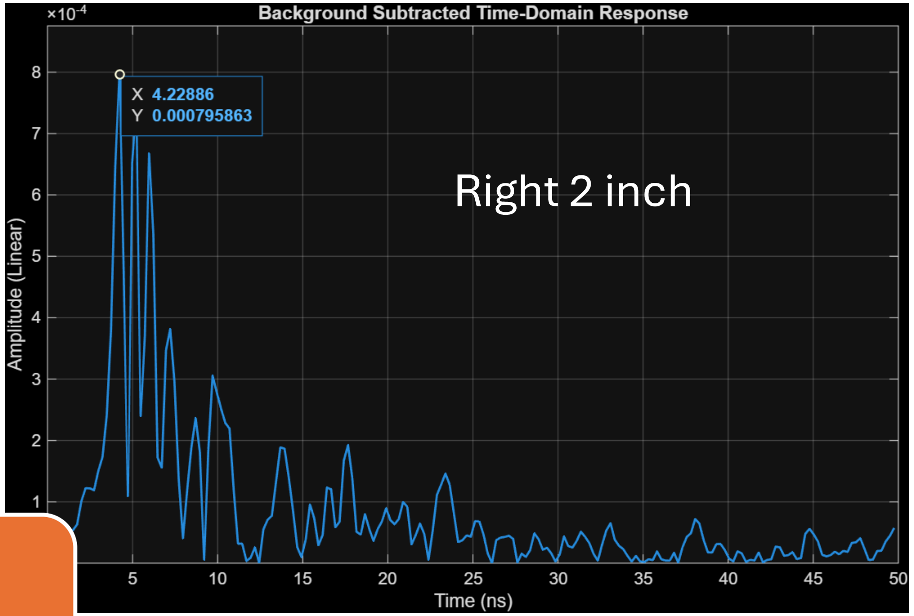
# Extended Study: Dual Target Detection



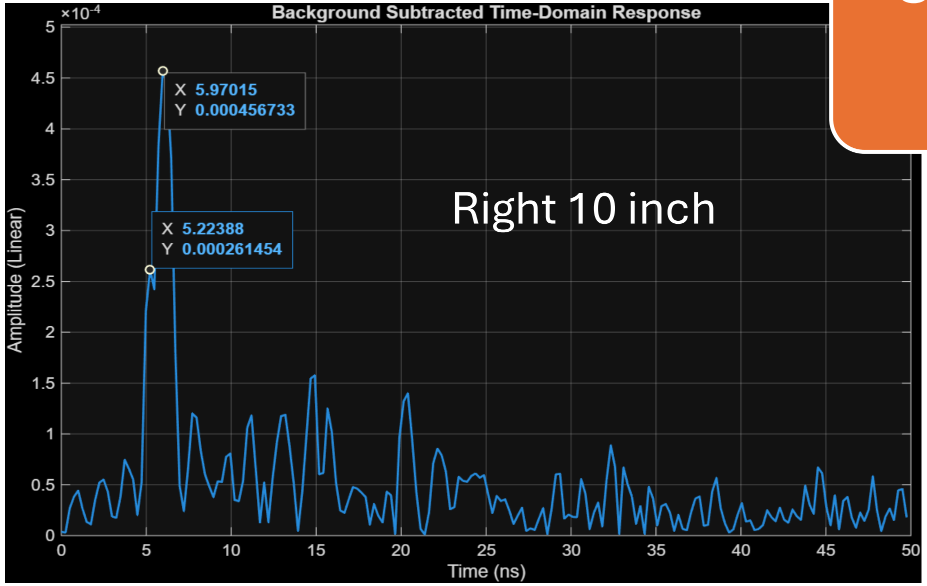

Two bottles used as targets

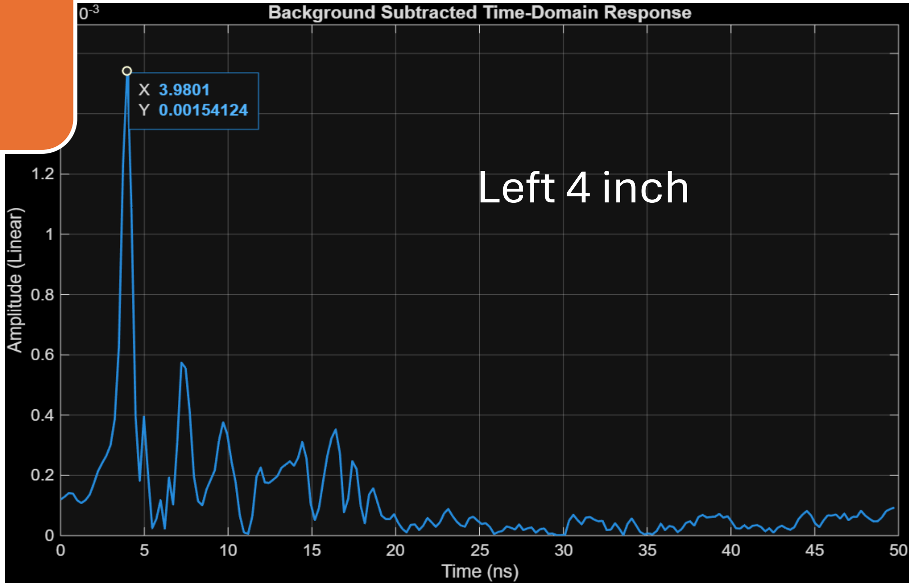
Cross-range scanning positions (left → center → right)

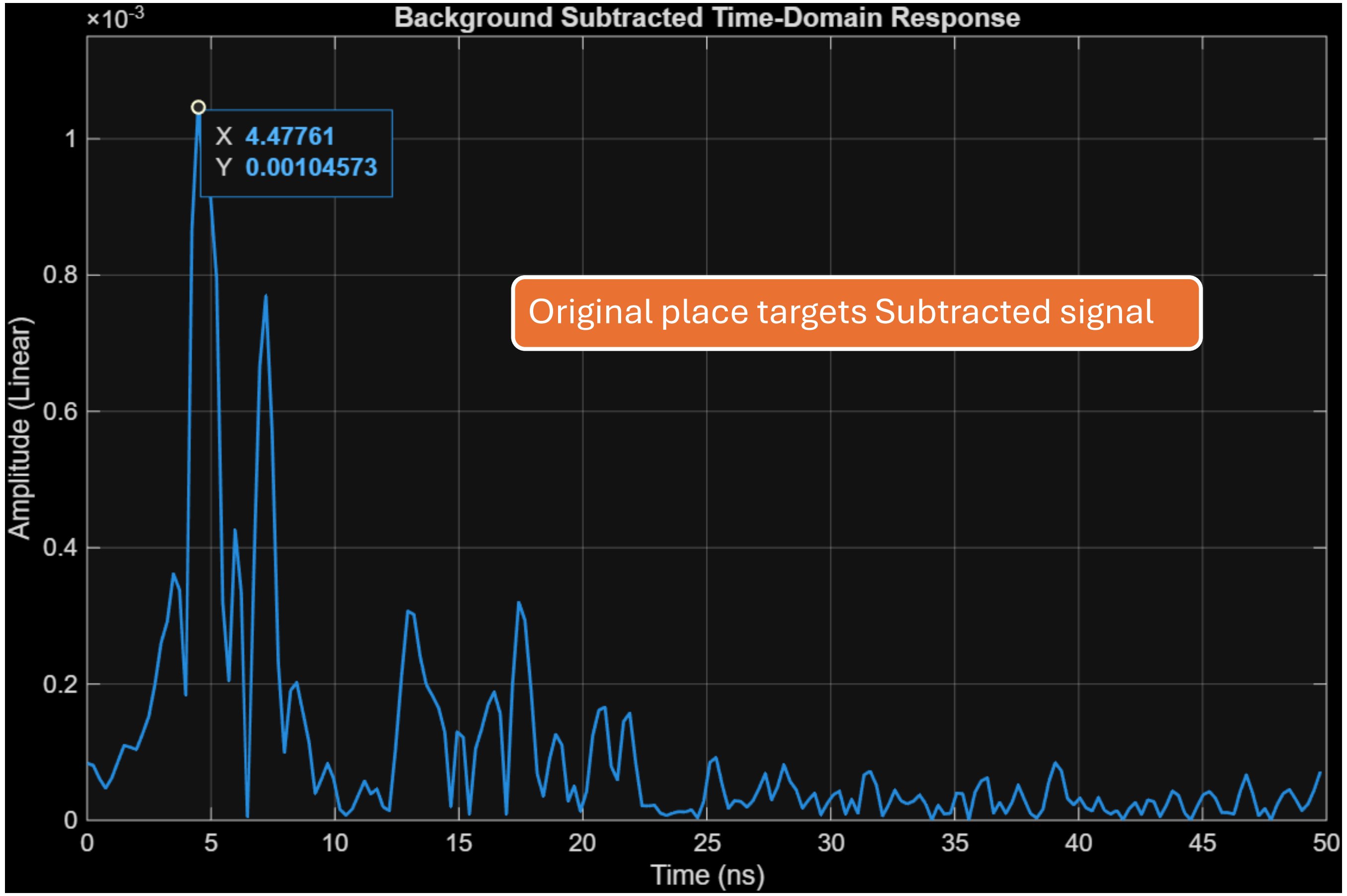



$$S\_diff(t) = S\_target(t) - S\_ref(t)$$






Transmitter to target 2 = 30 inch

**For 30 inches**

30 inch = **0.762 m**

$$\Delta t = \frac{2 \times 0.762}{3 \times 10^8} \approx 5.08\,\text{ns}$$

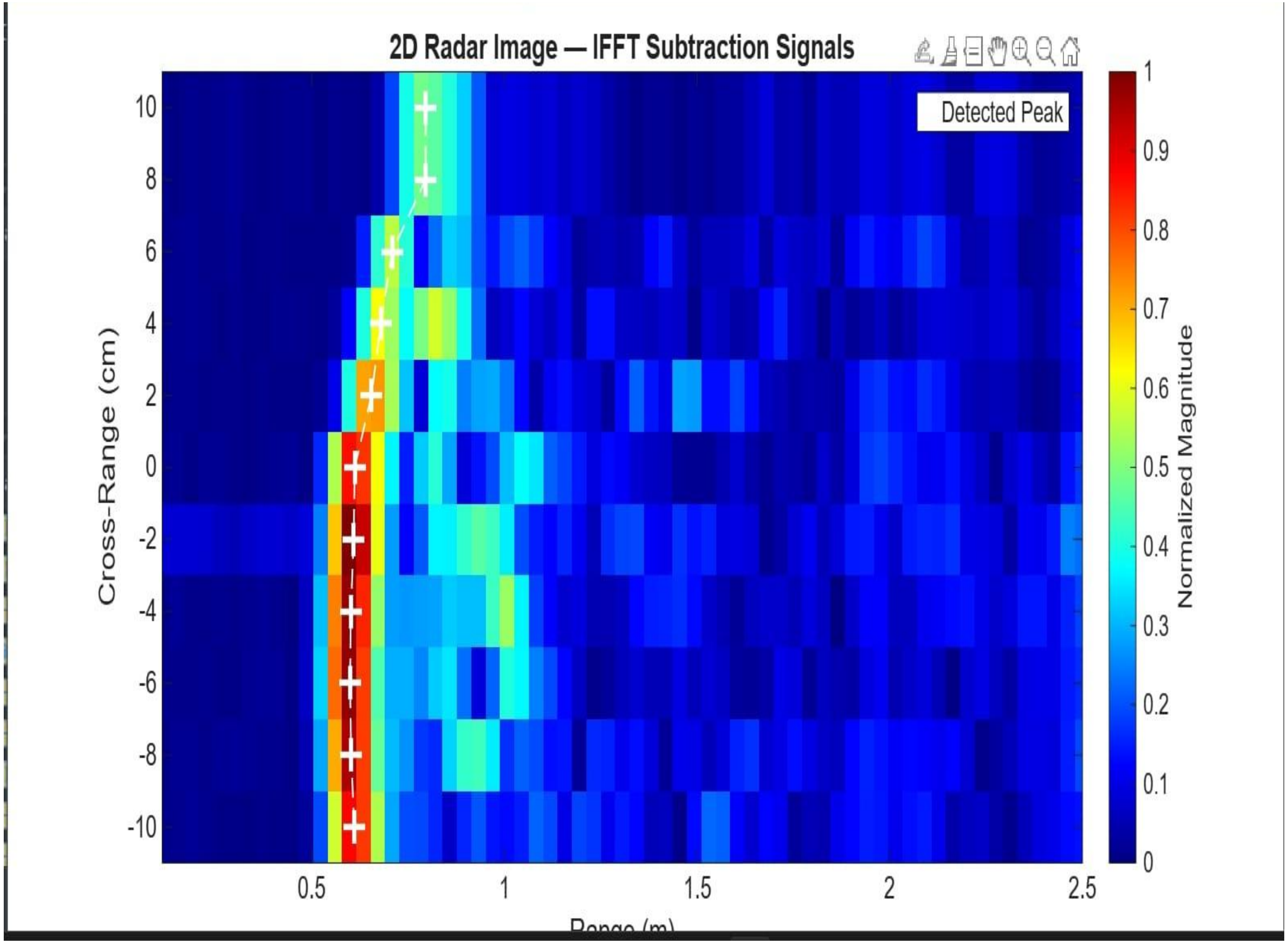
Transmitter to target 1 = 26 inch

**For 26 inches**

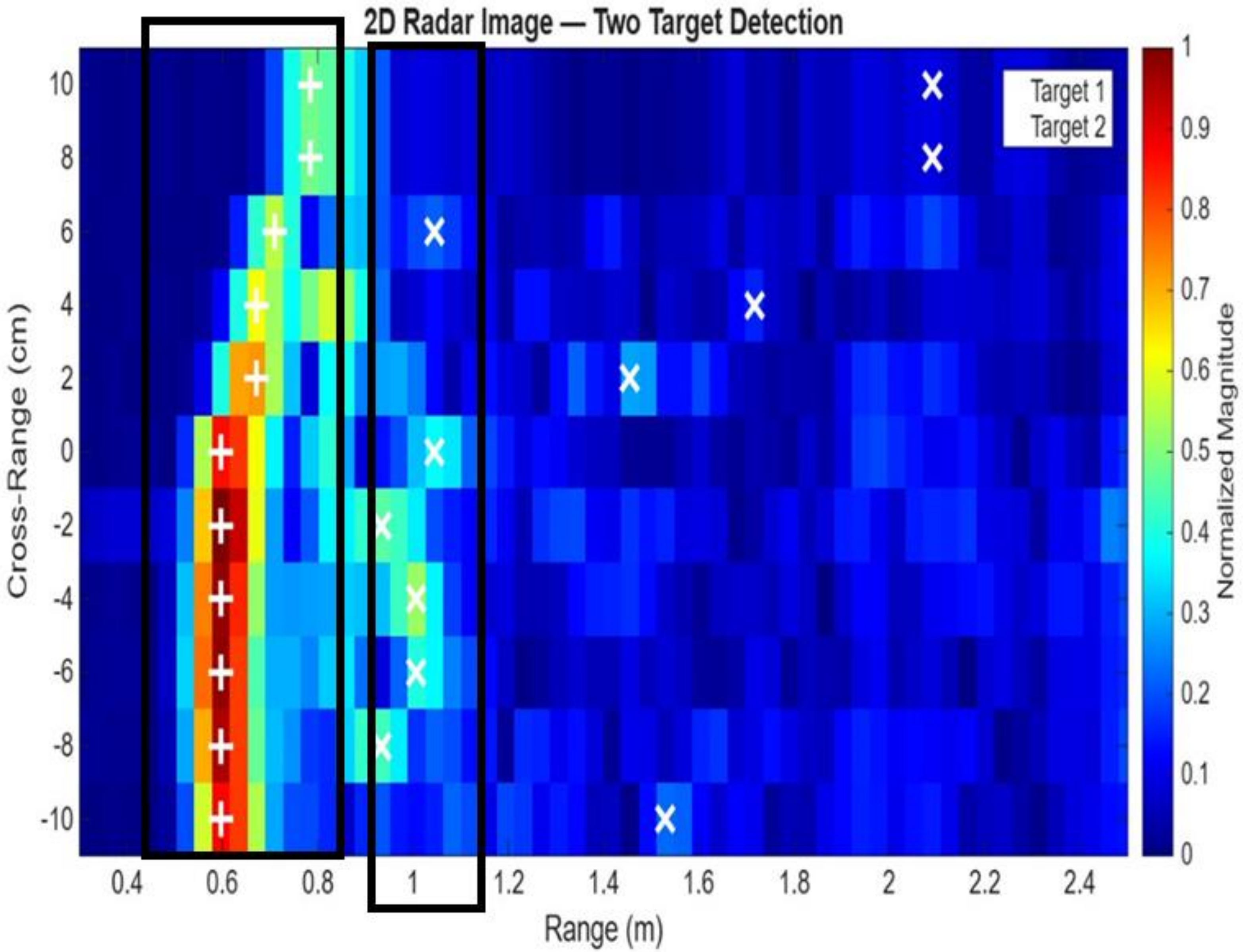
26 inch = **0.6604 m**

$$\Delta t = \frac{2 \times 0.6604}{3 \times 10^8} \approx 4.40\,\text{ns}$$

# 2D Heatmap of Target Reflections



After Reducing filtering, Additional reflection indication appear




Outline shows location of Targets for each position

# Conclusion

- A bistatic radar system for **target detection and distance estimation** was successfully implemented.
- Experimental results closely matched **theoretical time-of-flight calculations**.
- The system proved effective in detecting targets at different distances.
- The study demonstrates the feasibility of **low-cost RF-based radar sensing systems**.

# Thank you